

PATHFINDER.

BY PATHFINDER LIDAR SOLUTIONS

REQUIREMENTS

User Stories

70 user stories across 14 epics, with the Epic to Story to Criterion to Test traceability chain.

Project	Pathfinder Portal v2
Version	0.1 (Draft for review)
Date	2026-06-15
Owner	Pathfinder LiDAR Solutions

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Field	Value
Document	02 - User Stories
Version	0.1
Date	2026-06-15
Status	Draft
Source of truth	<code>docs/00-canonical-context.md</code>
Companion	<code>docs/03-acceptance-criteria.md</code>

Intro

This document enumerates the user stories for the Pathfinder portal, organized by the canonical epics E1 through E14. It draws all framing, terminology, identifiers, and product decisions from `docs/00-canonical-context.md` ; where the two disagree, the canonical context wins.

Persona set

Pathfinder is a multi-level white-label platform, so the personas span the operator that runs a deployment, the client orgs that are its tenants, the field crew that produces the data, the responders the data ultimately serves, and the engineers who build and operate the platform.

- **Operator admin** - runs a Pathfinder deployment under its own brand (Pathfinder LiDAR itself, ELS911, or a future reseller). Owns the brand profile, the user roster, API keys, and platform settings. Maps to the platform-wide admin role.
- **Staff member** - operator personnel (project lead, mapper, surveyor) who scan facilities, build maps, run the review loop, and publish deliverables. Maps to the staff role; the surveyor is the field-capture specialization of this persona.
- **Client reviewer** - a named user inside a client org (school district safety director, 911 center supervisor, university facilities lead) who reviews, comments on, and approves the work scoped to their org. Maps to the org-scoped client role.
- **First responder** - law, fire, or EMS personnel who consume published maps and scans for orientation and pre-incident planning, often via low-friction or shared access.
- **Compliance officer** - operator or client-side role responsible for the audit trail, NG911/NENA export conformance, and the trust posture for gov/edu procurement.
- **Platform engineer** - the developer/operator persona who builds, tests, and ships the platform and owns the quality and delivery harness.

Traceability chain

Every story maps into the canonical chain **Epic -> US -> AC -> Test -> Sprint**. Each user story carries an ID `US-<epic>.<n>` . Each story is validated by one or more acceptance criteria `AC-<story>.<n>` in `docs/03-acceptance-criteria.md` , and each AC maps 1:1 to a Vitest/Playwright test `T-<AC>` in the forthcoming TDD plan. Stories cite the research briefs in `docs/research/` where a behavior is grounded.

E1 - White-Label Brand Layer

US-1.1 - Default Pathfinder brand

As an operator admin, I want the platform to render under the Pathfinder LiDAR brand by default, so that an unconfigured deployment still looks finished and on-brand. Notes/grounding: canonical branding model, default brand = Pathfinder LiDAR; dark slate palette.

US-1.2 - Drop-in brand swap

As an operator admin, I want to switch the active brand by setting `BRAND=els911` and dropping in a logo, so that I can rebrand the whole deployment without editing components. Notes/grounding: canonical "rebrand = add/swap one profile + drop in a logo; no component edits".

US-1.3 - Ready-made ELS911 profile

As an operator admin, I want ELS911 to ship as a ready-made brand profile, so that the flagship operator deployment is available out of the box. Notes/grounding: canonical ELS911 palette (red #B22234 / navy #002868) ported from v1 tokens.css.

US-1.4 - Token-only component theming

As a platform engineer, I want every component to reference brand tokens and never hard-code a brand color, so that a brand swap propagates everywhere with zero component edits. Notes/grounding: canonical "no hard-coded brand colors anywhere in components".

US-1.5 - Brand continuity with marketing site

As an operator admin, I want the logged-in product brand to match the marketing site, so that clients do not experience a jarring brand break at sign-in. Notes/grounding: research 04 pitfall "brand discontinuity between marketing site and logged-in product".

E2 - Identity, Roles & Access

US-2.1 - Email/password login

As a client reviewer, I want to log in with my email and password over a JWT session, so that I can securely reach the work scoped to my org. Notes/grounding: canonical auth = JWT (HS256, Web Crypto), ported from v1.

US-2.2 - Role-based access control

As an operator admin, I want admin, staff, and client roles enforced on every route and action, so that users only see and do what their role permits. Notes/grounding: canonical RBAC admin/staff/client, org-scoped.

US-2.3 - Org-scoped data isolation

As a client reviewer, I want my access scoped to my own org's data, so that I never see another tenant's facilities or maps. Notes/grounding: research 04 "tenant hierarchy shapes data isolation and auth scoping".

US-2.4 - Audit-logged authentication events

As a compliance officer, I want logins, logouts, and access denials recorded in the audit log, so that I can demonstrate an auditable access trail during procurement. Notes/grounding: compliance brief "immutable exportable audit logs (who viewed/edited/exported, when)".

US-2.5 - Forward-looking MFA/SSO

As an operator admin, I want the identity layer designed so MFA and SSO can be added later, so that we can meet stricter tenant security requirements without re-architecting auth. Notes/grounding: canonical "MFA/SSO forward-looking"; compliance brief expected controls.

E3 - Org Hierarchy & Dashboards

US-3.1 - Navigate the org hierarchy

As a staff member, I want to navigate Org -> District -> Facility -> Building, so that I can drill from a customer down to a specific structure. Notes/grounding: canonical hierarchy; research 04 "cap nesting at three levels".

US-3.2 - Breadcrumb switcher dropdowns

As a client reviewer, I want each breadcrumb segment to double as a dropdown of its siblings, so that I can jump between buildings without backing all the way out. Notes/grounding: research 04 standout pattern "breadcrumb segments double as dropdown switchers".

US-3.3 - Roll-up dashboards

As an operator admin, I want each hierarchy level to show a roll-up dashboard of the levels beneath it, so that I can see facility and project status at a glance from the top. Notes/grounding: canonical "roll-up dashboards"; research 04 "clear project-progress visibility".

US-3.4 - Designed empty states

As a client reviewer, I want an empty facility to show a sample preview and one clear first action, so that I understand what a finished deployment looks like. Notes/grounding: research 04 "empty states are a high-leverage onboarding moment".

US-3.5 - Favorites and saved views

As a staff member, I want to star facilities and buildings I work on often, so that I can reach them quickly without re-traversing the hierarchy. Notes/grounding: research 04 "saved views/favorites (manual starring), progressive disclosure".

E4 - Project & Review Workflow

US-4.1 - Create a project under a building

As a staff member, I want to create a project tied to a specific building with members and a status, so that work is tracked against the right structure. Notes/grounding: canonical `projects.building_id` FK; "projects, status, progress, members".

US-4.2 - Track project progress and status

As a client reviewer, I want to see each project's status and progress, so that I know where my deliverable stands without emailing the operator. Notes/grounding: research O4 "clients expect clear project-progress visibility".

US-4.3 - Version review loop

As a staff member, I want to upload V1, have the client notified, collect their annotations, revise, and publish V2, so that revisions follow a clear, repeatable loop. Notes/grounding: research O4 version loop "staff uploads V1 -> client notified -> client annotates -> staff revises -> V2 -> approval".

US-4.4 - Record an approval

As a client reviewer, I want to approve a specific version and have that approval logged, so that sign-off is explicit and auditable. Notes/grounding: research O4 "role-based approvals; logged auditable review trail".

US-4.5 - Manage project membership

As an operator admin, I want to add and remove members on a project, so that only the right staff and client users can act on it. Notes/grounding: canonical `project_members` ; RBAC org-scoped.

E5 - 2D Floorplan Annotation Engine

US-5.1 - View a floorplan PDF

As a staff member, I want to open a floorplan PDF in the in-portal viewer, so that I can inspect and mark it up without leaving the platform. Notes/grounding: canonical ported 2D engine PDF.js + canvas (v1 viewer.js ~196KB).

US-5.2 - Use the annotation tools

As a staff member, I want the 12+ ported annotation tools (rectangle, freehand, text, arrows, measurements), so that I can mark up floorplans with the proven v1 toolset. Notes/grounding: canonical "12+ annotation tools" ported from v1 viewer.js.

US-5.3 - Persist annotations as JSON

As a staff member, I want my annotations saved and reloaded exactly as drawn, so that work survives reloads and is portable. Notes/grounding: canonical E5 "JSON export"; v1 `annotations` table retained.

US-5.4 - Layered per-audience annotations

As a client reviewer, I want to toggle annotation layers by audience (responder / staff / facilities), so that each viewer sees only the relevant markings. Notes/grounding: competitive brief "layered map model with per-audience toggles".

US-5.5 - Client annotation in the review loop

As a client reviewer, I want to annotate the staff's draft directly, so that my feedback lands on the exact spot I mean. Notes/grounding: research O4 example story "comment on a specific room before sign-off".

E6 - Safety Mapping & NFPA Export

US-6.1 - Place NFPA 170 markers

As a staff member, I want to place safety markers (AED, exits, extinguishers, shutoffs) from the NFPA 170 catalog, so that map semantics are standardized and not freehand. Notes/grounding: compliance brief "locked symbol semantics (place from catalog, not freehand)".

US-6.2 - Auto-generated NFPA legend

As a staff member, I want the legend to auto-generate as a vertical list with hallway chips, so that every exported map carries a standardized, readable key. Notes/grounding: memory `feedback_els911_legend_autofit` ; compliance brief auto-generated legend.

US-6.3 - North-locked printable map render

As a first responder, I want maps that are north-oriented, gridded, and printable, so that a hard copy survives a network outage during an incident. Notes/grounding: competitive brief "north-oriented, printable maps; hard copies must survive a network outage".

US-6.4 - Batch multi-page PDF export

As a staff member, I want to queue a batch export of all building maps into a correctly ordered multi-page PDF, so that I can hand off a complete map packet in one job. Notes/grounding: canonical `export_jobs` ; canonical "batch PDF export; multi-page print ordering".

US-6.5 - Z-axis floor labels

As a compliance officer, I want every map and marker tagged with its z-axis floor label, so that the data carries dispatchable-location floor context for NG911. Notes/grounding: compliance brief NENA-REQ-003 dispatchable location; `compliance_meta` z-axis labels.

US-6.6 - Marker placement fidelity

As a staff member, I want markers to render exactly where I place them with no jitter or drift, so that the exported map is spatially trustworthy. Notes/grounding: canonical "marker placement, jitter, print order" battle-tested v1 logic.

E7 - Unified Scan Library & Media

US-7.1 - Unified media asset library

As a staff member, I want one library holding floorplan PDFs, point clouds, splats, walkthrough videos, and reference images per facility, so that all scan media lives in one place. Notes/grounding: canonical

`media_assets` types; supersedes flat `documents.doc_type` .

US-7.2 - Large multipart upload

As a staff member, I want to upload large scan files via resumable R2 multipart upload, so that a multi-hundred-MB asset uploads reliably without timing out. Notes/grounding: canonical R2 multipart upload (`DOCUMENTS` binding); sprint S4.

US-7.3 - Asset versioning

As a staff member, I want each asset to carry a version with capture date and surveyor, so that I can track which scan is current after a renovation. Notes/grounding: canonical `media_assets.version` ,

`capture_date` , `surveyor` ; competitive field-verification workflow.

US-7.4 - Point-cloud cold archive

As an operator admin, I want master PLY point clouds archived to a cold R2 tier and never served to clients, so that we keep raw data without paying hot egress or shipping huge PLYs. Notes/grounding: canonical "master PLY archived cold in R2, never served"; 3D brief "do not ship PLY to clients".

US-7.5 - SPZ delivery format

As a first responder, I want scans delivered as compact SPZ, so that a scene loads fast on a field device instead of failing on a giant PLY. Notes/grounding: 3D brief "deliver SPZ" (~90% smaller, 500K scene ~11.8MB / ~1.1s load).

E8 - 3D Scan Viewer

US-8.1 - MP4 walkthrough player

As a first responder, I want to watch the mp4 walkthrough of a facility, so that I get zero-friction orientation on any device before entry. Notes/grounding: 3D brief priority 1 "MP4 walkthrough player ... ship first".

US-8.2 - Interactive splat viewer

As a first responder, I want to explore the Spark splat scene with orbit, fly, and walk camera modes, so that I can virtually walk a building I have never entered. Notes/grounding: canonical Spark (THREE.js, WebGL2); 3D brief camera modes.

US-8.3 - 3D measurement tool

As a staff member, I want to measure point-to-point distances and room dimensions in the scan, so that I can support staging and apparatus placement. Notes/grounding: 3D brief priority 3 measurement via Spark raycast-into-splat (differentiator).

US-8.4 - Anchored 3D markers

As a staff member, I want to drop markers (exits, hazards, shutoffs, AED) anchored in the 3D scene with a title, description, and saved viewpoint, so that key points are pinned in space. Notes/grounding: canonical `scan_markers_3d` ; 3D brief cap ~25/scene.

US-8.5 - Floor switching

As a first responder, I want to switch between per-floor scans, so that I can move through a multi-story school floor by floor. Notes/grounding: 3D brief priority 6 floor/level switching (essential for multi-story).

US-8.6 - Named viewpoints

As a staff member, I want to save named camera viewpoints, so that I can build a guided key-points tour of a scene. Notes/grounding: canonical `viewpoints` ; 3D brief priority 5 named bookmarks.

US-8.7 - Scan version compare

As a client reviewer, I want to compare scan versions by capture date, so that I can see what changed after a renovation. Notes/grounding: 3D brief priority 8 scan-version comparison.

E9 - Collaboration

US-9.1 - Anchored comment threads

As a client reviewer, I want to start a comment thread anchored to a specific map region or 3D marker, so that staff know exactly what I am referring to. Notes/grounding: research 04 "inline comment threads anchored to a specific region, with nested replies".

US-9.2 - Resolve workflow

As a staff member, I want to resolve a thread so it hides without deleting, so that handled feedback clears the view while the audit trail is preserved. Notes/grounding: research 04 "hide resolved comments, do not delete (preserves audit trail)".

US-9.3 - @-mentions with deep-link email

As a client reviewer, I want to @-mention a staff member and have them emailed a deep link, so that the right person is pulled straight to the comment. Notes/grounding: research 04 "@-mentions triggering targeted email notifications with a deep link".

US-9.4 - Batched activity notifications

As a staff member, I want my notifications batched into a digest, so that an active project does not flood my inbox. Notes/grounding: research 04 "batching to avoid notification fatigue"; pitfall "ungoverned notification volume".

US-9.5 - External share links

As an operator admin, I want to generate a scoped share link for a stakeholder outside the portal, so that I can share a map or scan without provisioning an account. Notes/grounding: research 04 "export/share links for stakeholders outside the portal".

E10 - Global Search

US-10.1 - Search across all entity types

As a staff member, I want one search box covering facilities, buildings, projects, documents, and marker labels, so that I can find anything by typing instead of navigating. Notes/grounding: canonical FTS5 across facilities/buildings/projects/documents/markers.

US-10.2 - Org-scoped search results

As a client reviewer, I want search results limited to my own org, so that I never see another tenant's entities in results. Notes/grounding: canonical org-scoped; research O4 data isolation.

US-10.3 - Search-centric navigation for large lists

As a staff member, I want search to be a primary way to reach a facility, so that large flat lists do not force me to browse the whole hierarchy. Notes/grounding: research O4 "lean on search-centric navigation for large flat lists".

E11 - Compliance & Trust

US-11.1 - NG911/NENA GeoJSON export

As a compliance officer, I want to export facility data as GeoJSON aligned to the NENA STA-006 schema, so that the data is usable by NG911 PSAP systems. Notes/grounding: compliance brief "export in NENA-aligned GIS formats (GeoJSON to STA-006 schema)".

US-11.2 - Compliance metadata per facility

As a compliance officer, I want per-facility fields for last-reviewed date, last responder tour, mandate flags, and drill links, so that I can demonstrate current, mandated mapping. Notes/grounding: canonical `compliance_meta` ; compliance brief Alyssa's/Kari's/HB3 metadata.

US-11.3 - Immutable, exportable audit log

As a compliance officer, I want an immutable audit log of who viewed, edited, and exported which map and when, exportable on demand, so that I can satisfy a security review. Notes/grounding: compliance brief immutable exportable audit logs; canonical `audit_log` .

US-11.4 - Public trust page

As an operator admin, I want a trust page stating our application-level controls and posture (FERPA/DPA, encryption, SOC 2 path), so that procurement is not disqualified for its absence. Notes/grounding: research O4 "a dedicated security/trust page is table stakes ... absence is a procurement disqualifier".

US-11.5 - Re-verification reminders

As a compliance officer, I want reminders when a facility's last-reviewed date goes stale, so that maps stay accurate and current as the mandate requires. Notes/grounding: compliance brief "re-verification reminders"; Alyssa's Law mapping mandate.

E12 - Accessibility

US-12.1 - Full keyboard navigation

As a client reviewer who does not use a mouse, I want to operate the portal entirely by keyboard, so that I can review and approve work without a pointing device. Notes/grounding: compliance brief WCAG 2.1 AA, keyboard-navigable controls.

US-12.2 - Screen-reader labels

As a client reviewer using a screen reader, I want every control and map symbol to have a descriptive label, so that I can understand the interface and the map non-visually. Notes/grounding: compliance brief "screen-reader labels on symbols and exports".

US-12.3 - Non-visual map alternative

As a first responder using assistive tech, I want a text list of rooms, exits, and markers as an alternative to the visual map, so that the map's information is available non-visually. Notes/grounding: compliance brief "accessible non-visual map alternatives (text room/exit lists)".

US-12.4 - Contrast and VPAT posture

As a compliance officer, I want the UI to meet WCAG 2.1 AA contrast and a maintained VPAT to be available, so that we clear ADA Title II and Section 508 procurement gates. Notes/grounding: compliance brief WCAG 2.1 AA min / 2.2 forward; VPAT/ACR; ADA Title II April 24 2026.

E13 - Admin & Platform Ops

US-13.1 - User management

As an operator admin, I want to create, edit, deactivate, and assign roles to users, so that I control who has access across the deployment. Notes/grounding: canonical E13 "user mgmt"; RBAC.

US-13.2 - API key management

As an operator admin, I want to issue, scope, and revoke API keys, so that integrations can authenticate without a user session and I can cut off a leaked key. Notes/grounding: canonical API keys; `api_keys` table; competitive "open API / export".

US-13.3 - Audit log viewer

As a compliance officer, I want an in-portal viewer to filter and review audit log entries, so that I can investigate access without exporting raw data first. Notes/grounding: canonical E13 "audit viewer"; `audit_log`.

US-13.4 - Platform settings

As an operator admin, I want a settings area for deployment-wide configuration, so that I can adjust platform behavior without a code change. Notes/grounding: canonical E13 "settings".

US-13.5 - Observability

As a platform engineer, I want platform observability (logs, errors, key metrics), so that I can detect and diagnose issues in the running deployment. Notes/grounding: canonical E13 "observability".

E14 - Quality & Delivery Harness

US-14.1 - Failing test before implementation

As a platform engineer, I want each acceptance criterion expressed as a failing test before code is written, so that the implementation is driven by the AC it satisfies. Notes/grounding: canonical TDD; AC -> Test T-<AC> traceability.

US-14.2 - Engine unit tests

As a platform engineer, I want each ported engine (2D annotation, NFPA export) to have its own Vitest unit tests as it is extracted, so that the battle-tested v1 logic is locked down. Notes/grounding: canonical "each engine gets its own unit tests as it is extracted".

US-14.3 - Playwright screenshot assertions

As a platform engineer, I want Playwright to capture and compare screenshots of UI-visible outcomes, so that visual regressions are caught automatically. Notes/grounding: canonical E14 "Vitest + Playwright screenshots".

US-14.4 - CI on every change

As a platform engineer, I want the full test suite to run in CI on every change, so that no regression merges unnoticed. Notes/grounding: canonical E14 "CI".

US-14.5 - Deploy to isolated CF test project

As a platform engineer, I want deploys to go to the `pathfinder` Pages project via `wrangler pages deploy`, so that testing never touches els911-portal production. Notes/grounding: canonical deploy direct; test project `pathfinder.pages.dev` isolated; memory CF direct-deploy.

Story index

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US-1.1	Default Pathfinder brand	E1	Must
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US-11.4	Public trust page	E11	Should
US-11.5	Re-verification reminders	E11	Could
US-12.1	Full keyboard navigation	E12	Must
US-12.2	Screen-reader labels	E12	Must
US-12.3	Non-visual map alternative	E12	Should
US-12.4	Contrast and VPAT posture	E12	Must
US-13.1	User management	E13	Must
US-13.2	API key management	E13	Must
US-13.3	Audit log viewer	E13	Should
US-13.4	Platform settings	E13	Should
US-13.5	Observability	E13	Should
US-14.1	Failing test before implementation	E14	Must
US-14.2	Engine unit tests	E14	Must
US-14.3	Playwright screenshot assertions	E14	Must
US-14.4	CI on every change	E14	Must
US-14.5	Deploy to isolated CF test project	E14	Must

Total: 70 stories across 14 epics.